



Influence of surface acid and base sites on the Guerbet coupling of ethanol to butanol over metal phosphate catalysts



Sabra Hanspal, Zachary D. Young, J. Tyler Prillaman, Robert J. Davis *

Department of Chemical Engineering, University of Virginia, 102 Engineers' Way, Charlottesville, VA 22904-4741, USA

ARTICLE INFO

Article history:

Received 9 February 2017

Revised 14 April 2017

Accepted 27 April 2017

Available online 7 June 2017

Keywords:

Hydroxyapatite

Fluorapatite

Magnesia

Guerbet reaction

Ethanol

Butanol

ABSTRACT

Hydroxyapatite (HAP; $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$) is a well-recognized catalyst for the Guerbet coupling of ethanol to butanol. In an effort to explore the role of the anion components of the catalyst, steady-state, gas phase catalytic coupling of ethanol to butanol was investigated at 633 K and atmospheric pressure over beta tricalcium phosphate (β -TCP; β - $\text{Ca}_3(\text{PO}_4)_2$) and fluorine-substituted hydroxyapatite (FAP; $\text{Ca}_{10}(\text{PO}_4)_6\text{F}_2$). Both β -TCP and FAP catalysts were catalytically active for butanol formation, leading to ~35% selectivity at low conversion, suggesting that the PO_4^{3-} group contributes to the active acid-base site pair for butanol formation during ethanol coupling over HAP. Co-feeding water, a product of ethanol coupling, revealed weaker inhibition of the rate over HAP relative to MgO, confirming the potential negative influence of strong base sites on coupling catalysts. Catalytic reactions of ethanol over $\text{Mg}_3(\text{PO}_4)_2$, β -TCP, and $\text{Sr}_3(\text{PO}_4)_2$ catalysts demonstrated the importance of Lewis acidity of the metal phosphates on the reaction. Relatively strong Lewis acid sites on the $\text{Mg}_3(\text{PO}_4)_2$ surface (Mg^{2+} cations) favored undesired ethanol dehydration to ethene (36% selectivity) and diethyl ether (52% selectivity) whereas the $\text{Sr}_3(\text{PO}_4)_2$ catalyst predominantly catalyzed ethanol dehydrogenation to acetaldehyde (91% selectivity) at a rate significantly higher than that observed over the other catalysts. Evidently, the β -TCP exposes intermediate-strength Lewis acid sites provided by surface Ca^{2+} cations that enable the material to effectively convert ethanol to butanol with 35% selectivity.

© 2017 Elsevier Inc. All rights reserved.

1. Introduction

The catalytic upgrading of bioethanol into value-added fuels and chemicals, such as butanol, has attracted increased interest in recent years due to the many drawbacks posed by ethanol as a gasoline fuel additive. Butanol is a more desirable transportation blend fuel compared to ethanol because it has a lower vapor pressure, a lower solubility in water, and a lower corrosivity [1]. These properties allow butanol to be easily transported using the existing fuel distribution infrastructure and to be blended with gasoline at higher concentrations than ethanol. Additionally, the energy content of butanol is closer to that of conventional petroleum, which allows for improved gas mileage for biofuel blended gasoline [1,2].

The catalytic transformation of ethanol to butanol occurs via the Guerbet coupling reaction – a well-known industrial route for higher alcohol synthesis. A commonly proposed mechanism for Guerbet coupling consists of a multi-step sequence of reactions that involves an initial dehydrogenation step to form acetaldehyde, which then undergoes an aldol self-condensation reaction to cro-

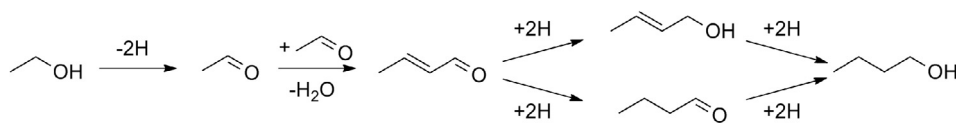
tonaldehyde, followed by hydrogenation of the condensation product to give butanol as shown in Scheme 1 [3,4]. A variety of catalytic materials have been studied for this reaction including solid base metal oxides (such as MgO) [5–7], Mg-Al mixed metal oxides [8–10], basic zeolites [11,12], and more recently hydroxyapatite [13–15].

While the mechanism in Scheme 1 is generally accepted for Guerbet coupling, other reaction paths have been proposed. One alternative mechanism involves the “direct coupling” of ethanol to butanol as suggested by Yang and Meng [12] for coupling over a basic zeolite. A direct coupling path has also been invoked by others [6,16]. Moreover, Scalbert et al. present a thermodynamic argument to conclude the aldol condensation of acetaldehyde is not kinetically relevant at high temperature (above 623 K) during alcohol coupling [15]. Evidently, additional mechanistic studies are needed for the Guerbet coupling reaction.

Calcium hydroxyapatite (HAP; $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$) has demonstrated exceptional catalytic performance in the Guerbet coupling of ethanol to butanol exhibiting unusually high activity and high butanol selectivity (70%) at significantly lower reaction temperatures than what has been previously reported in the literature for this reaction [17,18]. Recent work from our group has shown that

* Corresponding author.

E-mail address: rjd4f@virginia.edu (R.J. Davis).



Scheme 1. A commonly proposed mechanism of Guerbet coupling.

the HAP surface exposes a high surface density of acid and base sites of moderate strength that likely facilitate the reaction and account for its high catalytic activity and unique selectivity observed during the reaction [19,20]. However, the nature and composition of the active acid-base site pair on the HAP surface responsible for butanol formation has not been clearly assigned, which continues to complicate future catalyst design and optimization.

In a recent study by Ho et al. [21], *in situ* titration experiments combined with IR spectroscopy revealed the presence of two different active sites on the surface of HAP: surface Ca–O sites responsible for ethanol dehydrogenation to acetaldehyde and Ca^{2+} – PO_4^{3-} site pairs required for butanol formation. Diallo Garcia et al. [22] also investigated acid-base site pairs on the HAP surface and concluded strongly basic surface OH^- groups are likely the active base site involved in catalytic reactions because O^{2-} anions of surface PO_4^{3-} groups were weakly basic and unperturbed during acetylene adsorption studies. Clear consensus regarding the active acid-base site pairs on the HAP surface responsible for butanol formation is lacking and therefore additional investigations are needed.

In this work, the catalytic conversion of ethanol was investigated over beta-tricalcium phosphate (β -TCP; $\beta\text{-Ca}_3(\text{PO}_4)_2$) and fluorine-substituted hydroxyapatite (FAP; $\text{Ca}_{10}(\text{PO}_4)_6\text{F}_2$), and results were compared with those using stoichiometric HAP to gain insight into the importance of the OH^- and PO_4^{3-} anions of HAP. The influence of cations in the metal phosphate catalysts was also explored by investigating the catalytic activity of Mg- and Sr-phosphate in ethanol conversion. Additionally, to help elucidate the role of dihydrogen and water that are evolved in the Guerbet reaction, these by-products were co-fed with ethanol during the steady-state gas-phase conversion of ethanol over stoichiometric HAP.

2. Experimental methods

2.1. Catalyst preparation

Stoichiometric HAP was prepared using a controlled co-precipitation method based on the procedure reported by Tsuchida et al. [23], as described in detail in previous work [19]. Briefly, aqueous solutions of 0.5 M calcium nitrate tetrahydrate ($\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$, Acros Organics) and 0.3 M diammonium phosphate ($(\text{NH}_4)_2\text{HPO}_4$, Aldrich, >99.99%), previously adjusted to pH = 10.5 using aqueous ammonia, were simultaneously added dropwise to 100 cm^3 of distilled deionized water (DDI) held at 353 K. The resulting suspension was stirred for 24 h at 353 K under reflux. The precipitate was recovered by vacuum filtration, washed 3 times with DDI water, and dried in stagnant air at 400 K overnight. The final HAP powder was calcined at 873 K for 2 h in flowing air using a 10 K min^{-1} heating rate from room temperature and sized to 106–180 μm pellets prior to characterization and reactivity measurements.

The β -TCP catalyst was obtained by heating calcium-deficient hydroxyapatite (Sigma-Aldrich, CAS #7758-87-4, $\geq 98\%$) in flowing air to 1073 K using a thermal ramp rate of 2.5 K min^{-1} from room temperature and holding at 1073 K for 2 h. The phase transformation from calcium-deficient hydroxyapatite to β -TCP upon heating occurs according to the following reaction [24,25]:

$\text{Ca}_9(\text{HPO}_4)(\text{PO}_4)_5(\text{OH}) \rightarrow 3\text{Ca}_3(\text{PO}_4)_2 + \text{H}_2\text{O}$. The resulting powder was sized to 106–180 μm pellets prior to characterization and reactivity measurements.

The FAP catalyst was prepared by solid-state reaction [26]. A stoichiometric mixture (3:1 M ratio) of β -TCP (described above) and calcium fluoride (CaF_2 , Sigma-Aldrich) was heated using an electrical tube furnace in flowing argon (UHP, 99.999%) to 1373 K with a 4.5 K min^{-1} thermal ramp rate and holding at 1373 K for 4 h. Pure CaF_2 was placed in the furnace upstream of the reaction mixture to reduce the volatilization of fluorine [26,27].

Magnesium phosphate ($\text{Mg}_3(\text{PO}_4)_2$) was obtained by heating magnesium phosphate hydrate ($\text{Mg}_3(\text{PO}_4)_2 \cdot \text{H}_2\text{O}$, Sigma-Aldrich) in flowing air to 873 K using a thermal ramp rate of 10 K min^{-1} from room temperature and holding at 873 K for 2 h. The resulting $\text{Mg}_3(\text{PO}_4)_2$ powder was sized to 106–180 μm pellets prior to characterization and reactivity measurements. Strontium phosphate ($\text{Sr}_3(\text{PO}_4)_2$, Sigma-Aldrich) and magnesium oxide (MgO , Ube Material Industries, Ltd) were obtained commercially. The strontium phosphate and magnesium oxide powders were calcined at 873 K for 2 h in flowing air with a 10 K min^{-1} ramp rate. The resulting MgO powder was sized to 106–180 μm pellets prior to characterization and reactivity measurements.

2.2. Catalyst characterization

Crystalline phases of the catalytic materials were confirmed by powder X-ray diffraction (XRD) on a PANalytical X'Pert Pro Multi-Purpose Diffractometer using monochromatic $\text{Cu-K}\alpha$ radiation ($\lambda = 1.54 \text{ \AA}$). Scans were collected at $2\theta = 10\text{--}100^\circ$ with a 0.05° step size. Specific surface areas were obtained by N_2 adsorption at 77 K with a Micromeritics ASAP 2020 automated analyzer using the BET method after evacuation at 723 K for 4 h. Thermogravimetric analysis (TGA) experiments were conducted using a TA Instruments SDT Q600. Each sample was heated to 1273 K from room temperature with a 10 K min^{-1} thermal ramp rate under 100 $\text{cm}^3 \text{ min}^{-1}$ of flowing He.

Elemental analyses of the catalytic materials were performed by Galbraith Laboratories, Inc. (Knoxville, TN). The metal and phosphorus content of the catalysts were measured using inductively coupled plasma optical emission spectroscopy (ICP-OES) and the fluorine content of the fluorapatite sample was determined using a fluoride ion specific electrode. The surface composition of the hydroxyapatite sample was analyzed by X-ray photoelectron spectroscopy (XPS) using a Thermo Fisher ESCALab 250 apparatus. The signals were corrected using adventitious carbon ($\text{C}(1s)$) set to a binding energy of 284.6 eV.

The diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) of adsorbed pyridine was investigated using a Bio-Rad (FTS-60A) FTIR spectrometer equipped with a liquid-nitrogen-cooled MCT detector. The DRIFTS experiments were conducted using a high-temperature gas reaction chamber (Harrick Scientific) positioned onto a Praying Mantis diffuse reflectance sample accessory. All spectra were obtained by averaging 100 scans at a spectral resolution of 4 cm^{-1} . The magnesium phosphate sample, diluted in KBr by 50 wt.%, was loaded into the DRIFTS cell and thermally treated *in situ* at 773 K for 1 h in flowing He (30 $\text{cm}^3 \text{ min}^{-1}$) prior to pyridine exposure. Spectral backgrounds were then collected at 373 and 303 K as the sample cooled in flowing He. Pyridine

(Sigma-Aldrich) was introduced to the DRIFTS cell at 303 K for 1 min by passing He ($30 \text{ cm}^3 \text{ min}^{-1}$) through a pyridine saturator maintained at 273 K followed by purging under He flow ($30 \text{ cm}^3 \text{ min}^{-1}$) for 20 min. The catalyst sample was then heated to 373 K and IR spectra were collected after 15 min of holding at 373 K.

The nature, density, and strength of surface base and acid sites were characterized using adsorption microcalorimetry of carbon dioxide (CO_2) and triethylamine (TEA), respectively. The experiments were conducted at 303 K using a heat-flow microcalorimeter. Experimental procedures [7] as well as a detailed description of the apparatus [28] used for the calorimetry measurements have been reported in previous work. The samples were first outgassed at 773 K for 16 h under vacuum to a pressure below 10^{-3} Torr and then cooled to room temperature. Incremental doses of the gas probe molecule were introduced to the catalyst via a volumetric dosing system and each dose was allowed to equilibrate with the sample for 15 min. Adsorption isotherms and differential enthalpies of adsorption were obtained by measuring the amount of adsorbed species on the catalytic surface and the heats evolved for each dose.

2.3. Catalytic reactions

Catalytic reactions of gas-phase ethanol were conducted using a fixed-bed, stainless steel, tubular reactor (ID: 0.46 cm). The catalyst pellets were supported on a quartz wool plug in the reactor tube with a K-type thermocouple positioned at the center of the catalyst bed. Prior to reaction the catalysts were thermally treated *in situ* at 773 K for 1 h in flowing N_2 ($100 \text{ cm}^3 \text{ min}^{-1}$) after a 10 K min^{-1} thermal ramp. All catalytic experiments were performed at atmospheric total pressure. Anhydrous ethanol (Aldrich, 99.99%) was introduced to the reactor system by passing a N_2 carrier gas stream through a room temperature ethanol saturator. The reactor effluent streams were analyzed using an online Agilent 7890A gas chromatograph equipped with a Varian CP-PoraPLOT Q-HT column ($25 \text{ m} \times 0.32 \text{ mm} \times 10 \text{ }\mu\text{m}$) and a flame ionization detector.

Water co-feeding experiments were conducted in the same manner described above with liquid water fed by a syringe pump to a vaporizer connected to the reactor system. In kinetic studies where the amount of ethanol in the gas stream was varied, ethanol was fed to the vaporizer with a syringe pump.

The peak areas of reactants and products identified by GC were used to determine the ethanol conversion and selectivity of products. The conversion of ethanol was calculated as follows:

$$\text{conversion (C\%)} = \left(\frac{\sum n_i M_i}{2M_0} \right) \times 100$$

where n_i is the number of carbon atoms in product i , M_i is the molar flow rate of product i , and M_0 is the initial molar flow rate of ethanol. The selectivity toward product i was calculated on the basis of the total number of carbon atoms in the product and is therefore defined as:

$$\text{selectivity (C\%)} = \left(\frac{n_i M_i}{\sum n_i M_i} \right) \times 100$$

Aldol condensation reactions of acetaldehyde at 553 K were conducted using the same gas-phase reactor system described above. Acetaldehyde was introduced to the reactor system by passing a N_2 carrier gas stream through an acetaldehyde saturator maintained at 273 K with an ice-water bath. Reactions were carried out at 220 kPa total system pressure and 5 kPa acetaldehyde partial pressure. As severe catalyst deactivation was observed during aldol condensation, initial rates were obtained by fitting results to an empirical hyperbolic function and extrapolating to zero time on stream.

3. Results and discussion

3.1. Catalyst characterization

The BET surface areas of the catalytic materials studied in this work are listed in Table 1 along with the bulk metal and phosphorus content as determined by ICP-OES. The metal/phosphorus bulk molar ratio for all catalyst samples was close to the theoretical (stoichiometric) value. The fluorine content of the fluorapatite catalyst was measured to be 3.96 wt.%, slightly in excess of the theoretical value 3.77 wt.%. The measured chemical composition of the FAP sample (Ca:P:F = 10:6.1:2.2) was in good agreement with the theoretical stoichiometric composition for FAP (Ca:P:F = 10:6:2). Surface analysis of the stoichiometric HAP catalyst by XPS revealed a lower Ca/P surface molar ratio (1.46) than that of the bulk material (1.66). This finding appears to be a general characteristic of the HAP material that is often reported in the literature [18,29,30].

The XRD patterns of the stoichiometric HAP, FAP, MgO, and $\text{Sr}_3(\text{PO}_4)_2$ materials are presented in Fig. 1. The diffraction patterns of the synthesized HAP (Fig. 1a) and FAP (Fig. 1b) powders are consistent with the known hexagonal dipyramidal crystal structure of the apatite framework (ICDD: 01-074-9780) and no additional crystalline phases were detected. The magnesia catalyst (Fig. 1c) had an XRD pattern that is characteristic of periclase MgO, the cubic form of magnesium oxide with no observable defects (ICDD: 00-004-0829). The diffraction pattern for the strontium phosphate catalyst (Fig. 1d) is in agreement with the rhombohedral crystal structure of phase-pure $\text{Sr}_3(\text{PO}_4)_2$ (ICDD: 00-024-1008).

The powder X-ray diffraction patterns of calcium-deficient HAP as received at room temperature and after calcination at 1073 K are shown in Fig. 2a and b, respectively. The XRD pattern of calcium-deficient HAP after calcination (Fig. 2b) is consistent with the reference pattern for the rhombohedral form of β -TCP (ICDD: 00-009-0169) and confirms that the HAP material transformed to phase-pure β -TCP upon heating to 1073 K.

Fig. 3 displays the X-ray diffraction patterns of the magnesium phosphate hydrate precursor as received at room temperature (Fig. 3a) and the $\text{Mg}_3(\text{PO}_4)_2$ sample after calcination at 873 K (Fig. 3b). The $\text{Mg}_3(\text{PO}_4)_2$ pattern is in agreement with the monoclinic crystal structure of $\text{Mg}_3(\text{PO}_4)_2$ (ICDD: 00-025-1373). Additionally, the inherently different XRD patterns of the two materials confirm that a structural transformation from magnesium phosphate hydrate to magnesium phosphate occurred upon heating and is related to the loss of structural water from the crystal lattice.

To further verify that the magnesium phosphate hydrate ($\text{Mg}_3(\text{PO}_4)_2 \cdot \text{H}_2\text{O}$) material fully transformed to magnesium phosphate after calcination at 873 K, thermogravimetric analysis (TGA) was performed. The TGA profile of $\text{Mg}_3(\text{PO}_4)_2 \cdot \text{H}_2\text{O}$ is presented in Fig. 4. Three major regions of weight loss were observed corresponding to derivative weight loss peaks at 472, 625, and 706 K. These peaks are likely attributed to the loss of physisorbed and structural water. Negligible weight loss was observed from

Table 1
Specific surface areas and compositions of catalysts.

Catalyst	Sample identifier	S_{BET} ($\text{m}^2 \text{ g}^{-1}$)	M/P ratio bulk ^a
MgO	MgO	35	–
$\text{Mg}_3(\text{PO}_4)_2$	$\text{Mg}_3(\text{PO}_4)_2$	21	1.51
$\beta\text{-Ca}_3(\text{PO}_4)_2$	$\beta\text{-TCP}$	5.2	1.46
$\text{Ca}_{10}(\text{PO}_4)_6\text{F}_2$	FAP	0.5	1.64
$\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$	HAP	35	1.66
$\text{Sr}_3(\text{PO}_4)_2$	$\text{Sr}_3(\text{PO}_4)_2$	1.5	1.47

^a Atomic ratio of alkaline earth metal (M) and phosphorus (P) determined by ICP-OES.

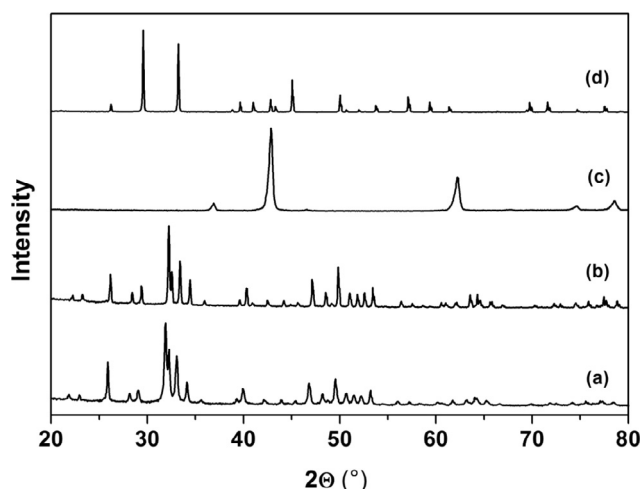


Fig. 1. X-ray diffraction patterns of (a) HAP, (b) FAP, (c) periclase MgO, and (d) strontium phosphate ($\text{Sr}_3(\text{PO}_4)_2$). Patterns are offset for clarity.

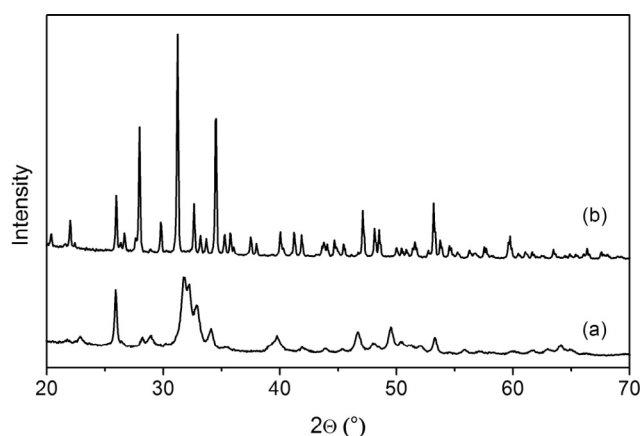


Fig. 2. Powder X-ray diffraction pattern of calcium-deficient hydroxyapatite at (a) room temperature and (b) after calcination at 1073 K for 2 h in flowing air to give β -TCP. Patterns are offset for clarity.

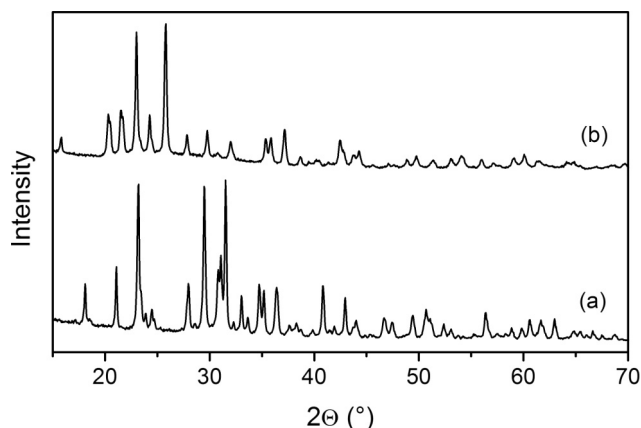


Fig. 3. Powder X-ray diffraction pattern of $\text{Mg}_3(\text{PO}_4)_2 \cdot \text{H}_2\text{O}$ at (a) room temperature and (b) after calcination at 873 K for 2 h in flowing air. Patterns are offset for clarity.

800 K to the final program temperature at 1273 K, implying that the final magnesium phosphate phase was fully dehydrated after calcination at 873 K.

The nature, density, and strength of surface base sites and acid sites of the catalytic materials were characterized using adsorption

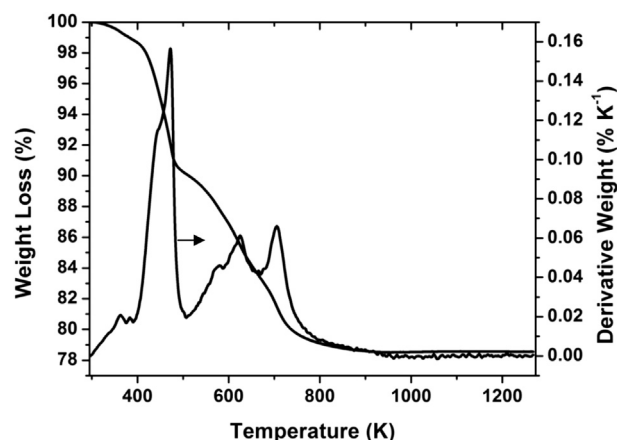


Fig. 4. Thermogravimetric analysis (TGA) profile of magnesium phosphate hydrate ($\text{Mg}_3(\text{PO}_4)_2 \cdot \text{H}_2\text{O}$).

microcalorimetry of CO_2 and TEA, respectively, at 303 K. Adsorption isotherms of CO_2 on HAP, MgO, $\text{Sr}_3(\text{PO}_4)_2$, and β -TCP are presented in Fig. 5a. Stoichiometric HAP exhibited the highest overall CO_2 uptake of $\sim 2.5 \mu\text{mol m}^{-2}$, which was calculated by extrapolating the physisorption region of the isotherm to zero pressure. The measured CO_2 adsorption capacities of the other catalysts were significantly lower, suggesting a much higher base site density exists on the surface of HAP compared to MgO, $\text{Sr}_3(\text{PO}_4)_2$, and β -TCP. The β -TCP catalyst had the lowest surface coverage of CO_2 ($\sim 0.3 \mu\text{mol m}^{-2}$) suggesting a small number of base sites present on this surface. The MgO and $\text{Sr}_3(\text{PO}_4)_2$ surfaces exhibited an intermediate base site density of $\sim 1.0 \mu\text{mol m}^{-2}$. The CO_2 adsorption isotherm on $\text{Sr}_3(\text{PO}_4)_2$ was measured over a smaller pressure range because of its very low surface area relative to the other materials (Table 1).

Differential heats of CO_2 adsorption as a function of surface coverage are presented in Fig. 5b. The highest initial heat of CO_2 adsorption was measured on the MgO surface ($-\Delta H_{\text{ads}} = 130 \text{ kJ mol}^{-1}$) suggesting MgO exposed the strongest base sites that

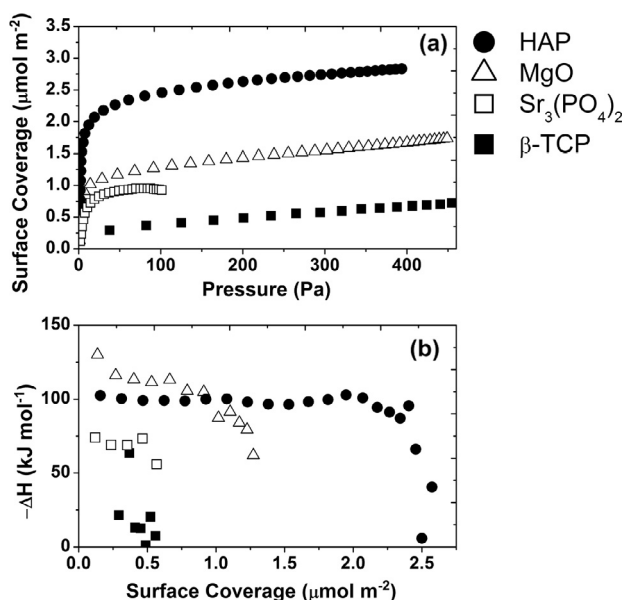


Fig. 5. Adsorption microcalorimetry of carbon dioxide on the catalytic materials at 303 K; (a) adsorption isotherms of carbon dioxide (b) differential heats of adsorption as a function of coverage.

interact with CO_2 relative to the other materials. Stoichiometric HAP possessed an intermediate-strength basicity with an initial heat of CO_2 adsorption of 102 kJ mol^{-1} . Low CO_2 adsorption enthalpies were measured on the metal phosphate catalysts, $\text{Sr}_3(\text{PO}_4)_2$, and $\beta\text{-TCP}$, which are indicative of lower strength base sites relative to HAP.

Infrared spectroscopy has revealed the presence of two adsorption sites on the surface of stoichiometric HAP for CO_2 : basic OH^- and O^{2-} of PO_4^{3-} groups [22]. These reports have shown that the phosphate groups are considerably less basic than the OH^- group and thus CO_2 predominantly interacts with surface hydroxyls on HAP, forming bicarbonate species [20]. These findings explain the lower CO_2 adsorption capacity and weaker base site strength measured on the metal phosphate catalysts, which lack structural hydroxyl groups, compared with HAP.

Fig. 6a shows the adsorption isotherms of TEA on the catalysts. The MgO catalyst displayed a linear isotherm for TEA uptake consistent with only physical adsorption. This implies that the MgO surface has a negligible density of Lewis acid sites on which to chemisorb TEA. All other catalysts had similar TEA uptakes ($1.7\text{--}1.9 \mu\text{mol m}^{-2}$) measured by extrapolating the physisorption portion of the isotherm back to zero pressure.

Whereas the phosphate catalysts have a similar number of TEA adsorption sites, the strength of the sites evaluated by the enthalpy of TEA adsorption varies with catalyst composition. The differential heat of adsorption on these materials can be seen in Fig. 6b. The HAP catalyst had the highest initial heat of adsorption of 135 kJ mol^{-1} while $\beta\text{-TCP}$ and $\text{Mg}_3(\text{PO}_4)_2$ had initial heats of adsorption of 101 and 112 kJ mol^{-1} , respectively. The initial heat of adsorption for $\text{Sr}_3(\text{PO}_4)_2$ could not be accurately measured in our system due to the very low surface area of the sample.

3.2. Catalytic reactions of ethanol

The reactions of ethanol over HAP were performed at 613 K. The total rate of ethanol conversion and product formation rates as a function of the initial ethanol concentration is presented in Fig. 7. All of the rates displayed in Fig. 7 were evaluated at similar ethanol conversion ($\sim 5\%$). The total rate of ethanol conversion, cal-

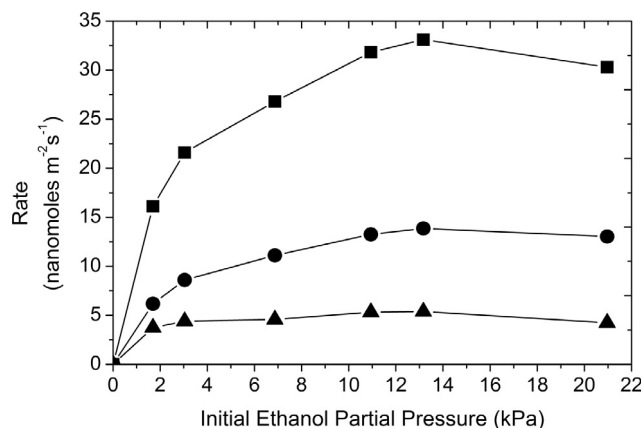


Fig. 7. The total rate of ethanol conversion (■) and the production rates of butanol (●) and acetaldehyde (▲) as a function of the initial ethanol concentration during the steady-state conversion of ethanol over stoichiometric calcium hydroxyapatite at 613 K and a total pressure of 1 atm.

culated as the sum of the product formation rates, appears to be approximately first-order at low initial ethanol concentrations and zero-order at higher concentrations of ethanol. This observation is consistent with results reported recently by Ho et al. [21].

3.2.1. Dihydrogen co-feeding experiments

One aspect of the Guerbet coupling of ethanol to butanol that remains unclear is the potential role of the hydrogen atoms that are produced in the reaction from alcohol dehydrogenation. There are several possible scenarios for the hydrogen: surface migration followed by recombination and desorption as dihydrogen, surface H-adatoms used for successive hydrogenation reactions of C-C coupling intermediates [31], and direct hydrogen transfer from alcohol to intermediates via an MPV-type reaction. To gain insight into the effect of gaseous H_2 on the Guerbet coupling of ethanol, H_2 was co-fed with ethanol during the steady-state reaction over stoichiometric hydroxyapatite at 613 K. The results obtained during the co-feed of H_2 to the reaction are presented in Fig. 8.

Increasing the concentration of gaseous dihydrogen, even to $>90\%$ of the feed, had no observable effect on the total rate of ethanol conversion or the butanol selectivity. Evidently, gaseous dihydrogen does not have a kinetic or thermodynamic role in the catalysis during Guerbet coupling of ethanol to butanol over

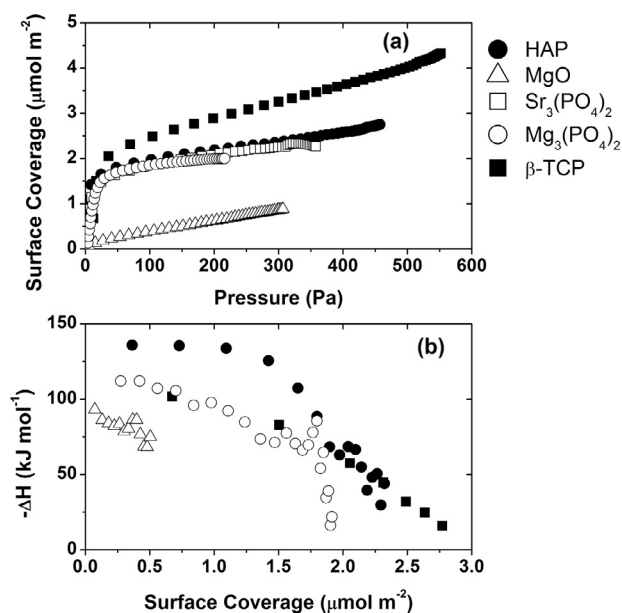


Fig. 6. Adsorption microcalorimetry of triethylamine on the catalytic materials at 303 K; (a) adsorption isotherms of triethylamine and (b) differential heats of adsorption as a function of coverage.

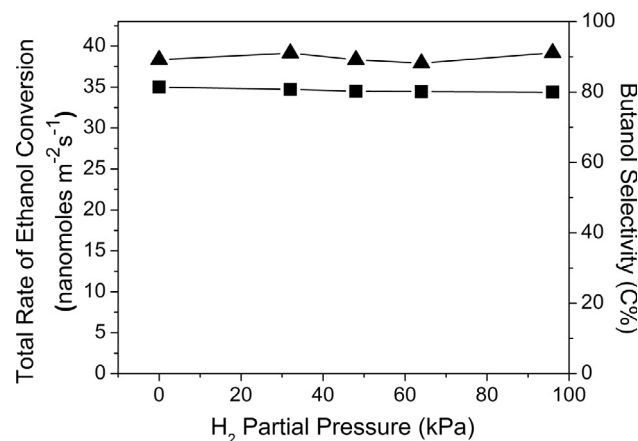


Fig. 8. The effect of increasing gas-phase dihydrogen on the total rate of ethanol conversion (▲) and the butanol selectivity (■) during the steady-state coupling of ethanol over stoichiometric hydroxyapatite at 613 K. The ethanol partial pressure was 6 kPa and a total pressure of 1 atm.

hydroxyapatite catalysts at the conditions of this study. This observation suggests that gas-phase dihydrogen is not the source of H atoms introduced in the hydrogenation steps in the Guerbet coupling reaction sequence over HAP. These results are consistent with those reported by Ogo et al. [13] who observed no hydrogenation products when crotyl alcohol was co-fed with H_2 over Sr-hydroxyapatite at 573 K. We suspect the hydrogen transfer reaction that forms product butanol originates from ethanol, presumably occurring via an MPV-type reaction. Additional studies of hydrogen transfer reactions are currently underway in our laboratory.

3.2.2. Water co-feeding experiments

Water is a by-product in the coupling of ethanol to butanol and therefore its influence on the overall reaction is of interest. The detrimental effect of water in liquid-phase systems has been reported and is generally ascribed to water accumulation in the reaction vessel that can lead to catalyst deactivation and/or undesirable side reactions (i.e. via base-catalyzed Tishchenko or Cannizzaro reactions) [32,33]. To the best of our knowledge, the role of water in the conversion of ethanol into butanol in the gas-phase has not been reported. Thus, the introduction and removal of water during the steady-state, vapor-phase conversion of ethanol was investigated over stoichiometric HAP at 613 K (Fig. 9) and MgO at 673 K (Fig. 10). The reaction with co-fed water was carried out at the same temperature at which the Guerbet coupling reaction was studied over each catalyst.

The introduction of water to the reaction over stoichiometric HAP rapidly inhibited production of butanol (Fig. 9). In contrast, the production rate of acetaldehyde was decreased by half of the steady-state rate prior to the addition of water. The removal of water from the feed to the reactor restored butanol and acetaldehyde formation rates to greater than 70% of their original steady-state values, suggesting weak and reversible interactions of water with the HAP surface. This inhibition of butanol formation is due to the co-fed water shifting the equilibrium of butanol formation and/or competitively adsorbing on catalytically active sites (Fig. 9).

Bolis and co-workers [34] investigated the hydrophilicity of HAP materials using adsorption microcalorimetry and IR spectroscopy measurements and found that water adsorption sites on the surface of HAP were composed of coordinatively unsaturated Ca^{2+} and nearby hydrophilic PO_4^{3-} species. The findings in that work

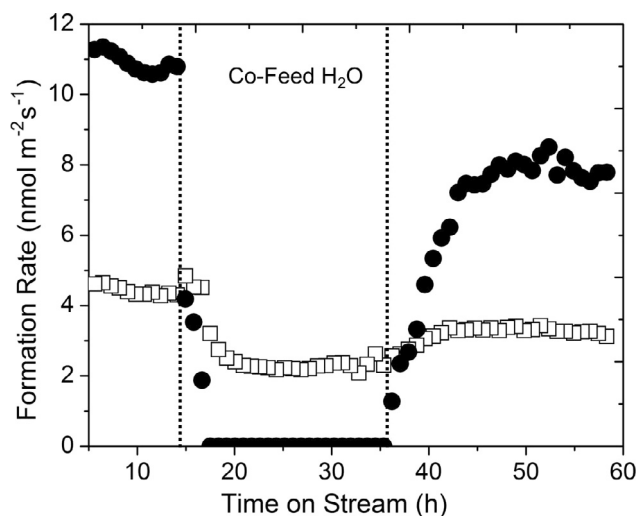


Fig. 9. Production rates of (●) butanol and (□) acetaldehyde obtained during the steady-state conversion of ethanol (6 kPa) over stoichiometric hydroxyapatite at 613 K and a total pressure of 1 atm with the addition and removal of water (6 kPa).

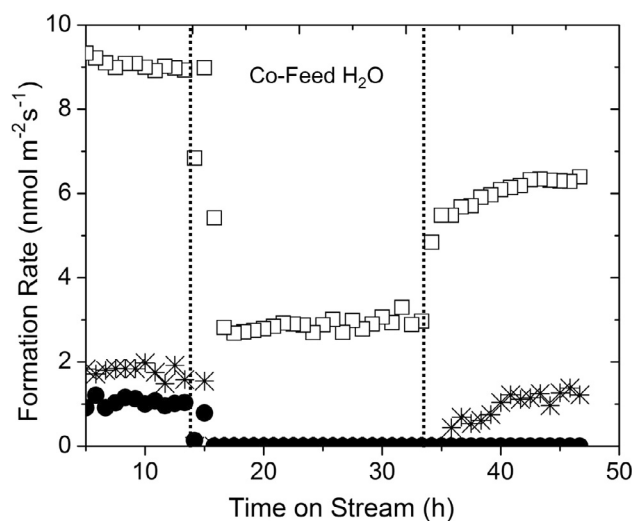


Fig. 10. Production rates of (●) butanol, (□) acetaldehyde, and (*) ethene obtained during the steady-state conversion of ethanol (6 kPa) over MgO at 673 K at a total pressure of 1 atm with the addition and removal of water (6 kPa).

are consistent with those reported recently by Diallo-Garcia et al. [22] who observed a decrease in the IR bands ascribed to O^{2-} of surface PO_4^{3-} groups upon water adsorption onto the HAP surface. The reported interaction of water with phosphate groups on the HAP surface together with the observed poisoning effect of water on butanol formation during the reaction (Fig. 9) suggests that the PO_4^{3-} group is likely involved in the active site for C–C bond formation during ethanol coupling over HAP. This finding is consistent with the reactivity results obtained during the ethanol coupling reaction over FAP and β -TCP discussed later (Table 2).

The effect of water co-fed with ethanol during the reaction over MgO at 673 K is presented in Fig. 10. The introduction of water decreased acetaldehyde formation rates by a factor of 3, whereas ethene and butanol production was eliminated after water addition. Similar to the results observed over stoichiometric HAP (Fig. 8), the presence of water during the reaction over MgO appears to have a greater inhibiting effect on C–C bond forming steps compared to ethanol dehydrogenation.

In contrast to the HAP surface, however, the active sites on MgO that participated in the production of butanol were not regenerated after stopping the addition of water. The sites on MgO responsible for ethene and acetaldehyde formation rates were nearly restored when water was removed from the feed. These observations suggest that water irreversibly adsorbed onto MgO surface sites that were catalytically active for butanol formation, most likely by dissociative adsorption. The dissociative adsorption of water on the MgO surface is well-studied and occurs on an acid-base site pair that contains strong base sites. The OH^- from water coordinates to a Lewis acid site (coordinatively unsaturated Mg^{2+} cation) and H^+ from water coordinates to a Brønsted base site provided by the O^{2-} anion [35,36]. Thus, based on these results it can be inferred that co-fed water dissociatively adsorbs onto Lewis acid-strong base site pairs (Mg^{2+} – O^{2-}) on MgO during the ethanol coupling reaction, which inhibits C–C bond formation. Interestingly, the irreversible adsorption of water onto these sites did not eliminate the production of acetaldehyde or ethene.

The base sites exposed on HAP are significantly weaker in strength compared to those on MgO, as measured by CO_2 adsorption microcalorimetry (Fig. 5b) and likely accounts for the weak and reversible surface interactions observed between water and the HAP surface during Guerbet coupling of ethanol. Additionally, the adsorption of water on HAP was explored by Corno et al. [37]

Table 2

Product distribution during ethanol coupling at 633 K.

Catalyst	Ethanol conv. (%)	Rate of ethanol conversion (mol m ⁻² s ⁻¹)	Rate of AcH production (mol m ⁻² s ⁻¹)	Rate of BuOH production (mol m ⁻² s ⁻¹)	Selectivity (C%)				
					Ethane	Ethene	AcH	DEE	BuOH
MgO	4.4	2.4×10^{-9}	1.4×10^{-9}	4.0×10^{-10}	0	8	59	0	33
Mg ₃ (PO ₄) ₂	5.0	6.5×10^{-9}	7.6×10^{-10}	0	0	36	12	52	0
β-TCP	4.2	2.7×10^{-8}	1.6×10^{-8}	4.7×10^{-9}	2	0	59	4	35
FAP	2.7	2.3×10^{-8}	1.4×10^{-8}	4.3×10^{-9}	0	0	62	0	38
HAP	4.9	8.7×10^{-8}	2.4×10^{-8}	3.1×10^{-8}	0	0	28	0	72
Sr ₃ (PO ₄) ₂	4.2	1.6×10^{-7}	1.5×10^{-7}	7.2×10^{-9}	0	0	91	0	9

AcH – acetaldehyde; DEE – diethyl ether; BuOH – butanol.

using density functional theory. They found that molecular adsorption of water occurs on the (001) surface of HAP with a heat of adsorption of approximately 80 kJ mol⁻¹ whereas dissociative adsorption of water occurs on the (010) HAP surface. The interaction of water was so strong on the (010) surface that the researchers speculate the surface was unlikely to be present on HAP that had been synthesized in an aqueous environment. This observation has important implications in reactions where water is formed *in situ* as a product such as Guerbet coupling and aldol condensation and may account for the significantly higher catalytic activity of HAP for these reactions compared to MgO [19,38].

3.2.3. Performance of phosphate catalysts

The observed inhibitory influence of water on butanol formation over stoichiometric HAP during the steady-state conversion of ethanol (Fig. 9) and prior observations that water adsorption on HAP occurs on PO₄³⁻ [22,34] motivated us to investigate the importance of the PO₄³⁻ group for C-C bond formation during the ethanol coupling reaction. In addition, the potential roles of cations (Ca²⁺, Mg²⁺, Sr²⁺) and anions (OH⁻, F⁻) on the reactivity of phosphate-based catalysts were explored. A variety of catalysts such as β-TCP, Sr₃(PO₄)₂, Mg₃(PO₄)₂, and FAP were tested in Guerbet coupling and compared to HAP. The product distribution and steady-state rate observed during ethanol conversion at 633 K for the catalysts are listed in Table 2. All catalytic reactions were conducted at low ethanol conversion (~5%) to ensure differential reactor conditions.

The HAP catalyst exhibited the highest catalytic activity for butanol production and the highest selectivity to butanol (72%) among all of the catalysts tested. The β-TCP and FAP catalysts were ~3 times less active on an area basis and ~50% less selective toward butanol, compared to HAP, which implies that the hydroxyl group of HAP likely plays a beneficial role in the catalysis during the coupling of ethanol to butanol. Although the catalytic activity and selectivity to butanol over β-TCP and FAP were not as impressive as those over HAP, the Guerbet coupling reaction was nevertheless still observed. This observation suggests that the PO₄³⁻ group is critical for the production of butanol over β-TCP and FAP and is likely involved in the active acid-base site pair for butanol formation during ethanol coupling over HAP. These results are consistent with the water co-feeding experiments over HAP (Fig. 9) where the inhibiting effect of water on butanol formation was likely the result of water interactions with adjacent Ca²⁺-PO₄³⁻ site pairs on the surface [34].

The MgO catalyst was the least active among all of the catalysts tested and catalyzed primarily the dehydrogenation of ethanol to acetaldehyde (59% selectivity). The undesired ethene, which was formed by dehydration of ethanol, was also observed (8% selectivity).

As discussed earlier, the Guerbet coupling of ethanol to butanol likely proceeds over an acid-base site pair, and the poor catalytic performance of MgO in alcohol coupling reactions has been attributed to strongly basic O²⁻ anions that are disproportionate in

strength and provide a low accessibility to the surface Lewis acid sites associated with coordinatively unsaturated cations (i.e. Mg²⁺) [9]. These strong base sites on MgO, as measured by CO₂ adsorption microcalorimetry (Fig. 5), also facilitate undesired ethanol dehydration to ethene via the E_{1CB} pathway [39]. The relative strength of the acid and base sites that are involved in the active site pair must be carefully balanced to achieve high activity and selectivity to butanol while minimizing unwanted by-product formation.

Increasing the acid site density on the MgO surface or balancing the relative strengths of the acid and base sites on the surface has been beneficial for the Guerbet coupling reaction [8,40]. This optimization of acid-base strength as a strategy for improving Guerbet coupling catalysts is consistent with the results presented in Table 2. The total rate of ethanol conversion over the β-TCP and FAP catalysts was an order of magnitude higher than that measured over MgO. Compared to MgO, these materials expose base sites of much weaker strength (Fig. 5b) and acid sites in significantly higher surface density (Fig. 6a). An increased number of modest-strength acid-base site pairs on the surface likely allows β-TCP and FAP to catalyze the ethanol conversion reaction at a significantly higher rate than on highly-basic MgO.

The critical requirement of balanced-strength acid and base sites for the Guerbet coupling reaction is clearly visible when comparing the results obtained during ethanol conversion over MgO and Mg₃(PO₄)₂. The phosphate group of Mg₃(PO₄)₂ is considerably less basic than the O²⁻ anions present on the MgO surface so that the catalysis shifts from ethanol dehydrogenation over MgO toward acid-catalyzed ethanol dehydration over Mg₃(PO₄)₂. The Mg₃(PO₄)₂ catalyst exhibited characteristic acid-like behavior leading to 52% and 36% selectivity toward diethyl ether and ethene, respectively. Additionally, this material was catalytically inactive for C-C coupling to butanol (Table 2).

To further characterize the acid sites present on the Mg₃(PO₄)₂ surface, DRIFTS of adsorbed pyridine was performed over a fresh catalyst diluted in KBr. Infrared spectroscopy of pyridine is commonly used to characterize acid sites because the IR signature of pyridine coordinated to a Lewis acid site on the surface is very different from that of the pyridinium ion, i.e. when it is associated with a Brønsted acid, which permits differentiation between acid types on solid acid surfaces [41]. It should be noted that the acid properties of the catalyst could be altered by exposure to reaction conditions (which produce water), so the conclusions from DRIFTS are relevant to the fresh catalyst.

The DRIFTS spectrum of adsorbed pyridine on Mg₃(PO₄)₂ at 373 K is presented in Fig. 11. Pyridine adsorbed on Mg₃(PO₄)₂ resulted in bands at 1608, 1576, 1491, and 1446 cm⁻¹ which are characteristic of Lewis-type coordination with the surface [42]. The absence of a band at 1540 cm⁻¹, characteristic of the pyridinium ion, indicates a lack of Brønsted acidity associated with the Mg₃(PO₄)₂ surface. Therefore, the surface acid sites exposed on Mg₃(PO₄)₂ are of the Lewis type and confirm that ethanol dehydration observed over this material was Lewis-acid catalyzed. The

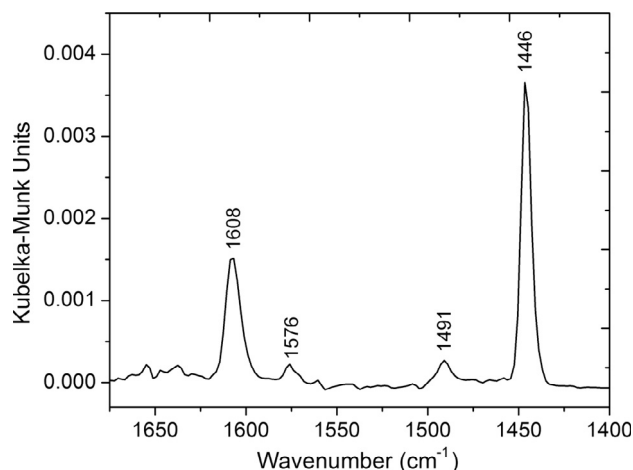


Fig. 11. DRIFTS of adsorbed pyridine on $\text{Mg}_3(\text{PO}_4)_2$ at 373 K after purging with He for 20 min. The observed peaks are characteristic of pyridine coordinately bonded to Lewis acid sites.

Lewis acid sites of $\text{Mg}_3(\text{PO}_4)_2$ dominated the catalyst performance favoring dehydration over coupling, whereas the basic sites of MgO allowed for coupling to occur. The $\text{Mg}_3(\text{PO}_4)_2$ seems to lack the appropriate strength acid-base site pairs needed for Guerbet coupling.

The influence of Lewis acidity on Guerbet coupling was explored by comparing the rates of ethanol reactions over $\text{Mg}_3(\text{PO}_4)_2$, β -TCP, and $\text{Sr}_3(\text{PO}_4)_2$. Decreasing the electronegativity of the metal cation from Mg^{2+} to Ca^{2+} significantly improved the catalytic performance for the coupling reaction. The β -TCP catalyst exhibited a higher rate of ethanol conversion, lower ethanol dehydration rate (0% selectivity to ethene), and higher selectivity to butanol (35% selectivity), relative to $\text{Mg}_3(\text{PO}_4)_2$. The acid-catalyzed bimolecular ethanol dehydration product, diethyl ether, was observed in minor quantities over β -TCP (4% selectivity), however, the acid-sites responsible for diethyl ether formation from primary alcohols have been shown to be weaker than those required for ethene formation, which is consistent with the expected decrease in Lewis acidity from Mg^{2+} to Ca^{2+} [8].

The rate of ethanol conversion over the $\text{Sr}_3(\text{PO}_4)_2$ catalyst was the highest of the pure phosphate materials (Mg^{2+} , Ca^{2+} , Sr^{2+}) and the product distribution was nearly all acetaldehyde (91% selectivity). Interestingly, the $\text{Sr}_3(\text{PO}_4)_2$ catalyst was significantly more active for ethanol dehydrogenation on a surface area basis than any of the other catalysts tested, including hydroxyapatite. The rate of acetaldehyde formation over $\text{Sr}_3(\text{PO}_4)_2$ was an order of magnitude higher than that observed over β -TCP and over two orders of magnitude higher than that observed over $\text{Mg}_3(\text{PO}_4)_2$. However, this material led to poor selectivity toward coupled products during the reaction (9% selectivity to butanol).

In an attempt to understand the coupling activity of $\text{Sr}_3(\text{PO}_4)_2$ we tested its catalytic activity in acetaldehyde condensation at 553 K with 5 kPa partial pressure of acetaldehyde, and compared the results to those obtained over anatase titania (TiO_2), HAP, and MgO (Table 3). The reaction over $\text{Sr}_3(\text{PO}_4)_2$ proceeded with 100% selectivity toward the condensation product, crotonaldehyde, with an initial rate higher than the other catalysts tested [38].

It is interesting that $\text{Sr}_3(\text{PO}_4)_2$ catalyzes aldol condensation quite readily but is a poor coupling catalyst under Guerbet coupling conditions. Related work in our laboratory has shown that a weak binding affinity for acetaldehyde on the titania surface is likely a favorable condition for aldol condensation, as it prevents successive coupling and rapid deactivation [38]. However, weak binding may also prevent C-C bond formation during Guerbet cou-

Table 3

Initial rate of aldol condensation of acetaldehyde (5 kPa) at 553 K.

Catalyst	Initial rate ($\text{mol m}^{-2} \text{s}^{-1}$)
$\text{Sr}_3(\text{PO}_4)_2$	2.8×10^{-7}
TiO_2^a	1.7×10^{-7}
HAP ^a	8.7×10^{-8}
MgO ^a	5.6×10^{-8}

^a Results were previously reported by Young et al. [38].

pling conditions at low partial pressures of acetaldehyde when ethanol can compete for surface active sites. It is possible that acetaldehyde has a weak binding affinity for the $\text{Sr}_3(\text{PO}_4)_2$ surface and therefore preferentially desorbs into the gas-phase during ethanol conversion, preventing subsequent C-C bond forming steps.

The weak interactions between acetaldehyde and the $\text{Sr}_3(\text{PO}_4)_2$ surface are related to the acid-base surface properties of the material. Adsorbed acetaldehyde produced during ethanol coupling may desorb or undergo base-catalyzed abstraction of the α -H to form an adsorbed surface enolate intermediate species that leads to longer-chain condensation products. However, if the base site (PO_4^{3-}) on $\text{Sr}_3(\text{PO}_4)_2$ is too weak for enolate formation, acetaldehyde will desorb. This scenario does not seem to be the case as higher CO_2 adsorption energies were measured on the $\text{Sr}_3(\text{PO}_4)_2$ surface from CO_2 adsorption microcalorimetry compared to those measured on β -TCP, which was catalytically active for coupling (Fig. 5). Thus it is likely that the poor coupling performance of $\text{Sr}_3(\text{PO}_4)_2$ compared to β -TCP is a result of weaker Lewis acid interactions between acetaldehyde and Sr^{2+} cations. This information is critical for the future development of effective catalytic materials for the Guerbet coupling reaction.

4. Conclusions

The β -TCP and FAP catalysts were not as effective as HAP for Guerbet coupling of ethanol, however the observed ~35% selectivity toward butanol indicates the phosphate group has adequate basicity for the reaction.

The introduction and removal of water vapor during the steady-state gas-phase conversion of ethanol over MgO revealed that water dissociatively adsorbs onto Lewis acid-strong base site pairs that were catalytically active for butanol formation. Compared to the MgO surface, the base sites exposed on HAP are significantly weaker in strength, as measured by CO_2 adsorption microcalorimetry and likely accounts for the reversible surface interactions observed between water and the HAP surface. This finding has important implications in the Guerbet coupling reaction during which water is formed as a product and may account for the significantly higher catalytic activity of HAP observed during the reaction compared to MgO.

The $\text{Mg}_3(\text{PO}_4)_2$ catalyst exhibited a product distribution characteristic of acid catalysts, leading to 52% and 36% selectivity toward diethyl ether and ethene, respectively. The $\text{Sr}_3(\text{PO}_4)_2$ catalyst was highly active and selective for ethanol dehydrogenation to acetaldehyde (91% selectivity) but exhibited poor selectivity toward C-C coupling products. Since $\text{Sr}_3(\text{PO}_4)_2$ is an excellent catalyst for acetaldehyde condensation, we attribute the poor selectivity of Guerbet coupling products to severe competitive adsorption of ethanol. Among the metal (Mg, Ca, Sr) phosphates, the β -TCP catalyst possessed an intermediate-strength Lewis acidity provided by surface Ca^{2+} cations that allow the material to effectively convert ethanol to butanol with 35% selectivity.

Acknowledgments

This work was supported by the Chemical Sciences, Geosciences and Biosciences Division, Office of Basic Energy Sciences, Office of Science, U.S. Department of Energy, grant no. DE-FG02-95ER14549. The authors would like to thank Dmitry Pestov at Virginia Commonwealth University for assistance with XPS and Dr. Gary Koenig's lab at the University of Virginia for help with the fluorapatite synthesis.

References

- [1] C. Jin, M. Yao, H. Liu, C.F. Lee, J. Ji, Progress in the production and application of n-butanol as a biofuel, *Renew. Sustain. Energy Rev.* 15 (2011) 4080–4106, <http://dx.doi.org/10.1016/j.rser.2011.06.001>.
- [2] N. Savage, Fuel options: the ideal biofuel, *Nature* 474 (2011) S9–S11.
- [3] A.J. O'Lenick, Guerbet chemistry, *J. Surf. Deterg.* 4 (2001) 311–315, <http://dx.doi.org/10.1007/s11743-001-0185-1>.
- [4] S. Veibel, J. Nielsen, On the mechanism of the Guerbet reaction, *Tetrahedron* 23 (1967) 1723–1733.
- [5] W. Ueda, T. Kuwabara, T. Ohshida, Y. Morikawa, A low-pressure guerbet reaction over magnesium oxide catalyst, *J. Chem. Soc., Chem. Commun.* (1990) 1558–1559.
- [6] A.S. Ndou, N. Plint, N.J. Coville, Dimerisation of ethanol to butanol over solid-base catalysts, *Appl. Catal. A-Gen.* 251 (2003) 337–345, [http://dx.doi.org/10.1016/S0926-860X\(03\)00363-6](http://dx.doi.org/10.1016/S0926-860X(03)00363-6).
- [7] T.W. Birky, J.T. Kozlowski, R.J. Davis, Isotopic transient analysis of the ethanol coupling reaction over magnesia, *J. Catal.* 298 (2013) 130–137, <http://dx.doi.org/10.1016/j.jcat.2012.11.014>.
- [8] J.I. Di Cosimo, V.K. Díez, M. Xu, E. Iglesia, C.R. Apesteguía, Structure and surface and catalytic properties of Mg–Al basic oxides, *J. Catal.* 178 (1998) 499–510, <http://dx.doi.org/10.1006/jcat.1998.2161>.
- [9] J. Di Cosimo, C. Apesteguía, M. Ginés, E. Iglesia, Structural requirements and reaction pathways in condensation reactions of alcohols on Mg₂AlO₇ catalysts, *J. Catal.* 190 (2000) 261–275, <http://dx.doi.org/10.1006/jcat.1999.2734>.
- [10] M. León, E. Díaz, S. Ordóñez, Ethanol catalytic condensation over Mg–Al mixed oxides derived from hydrotalcites, *Catal. Today* 164 (2010) 436–442.
- [11] K. Gotoh, S. Nakamura, T. Mori, Y. Morikawa, Supported alkali salt catalysts active for the guerbet reaction between methanol and ethanol, *Stud. Surf. Sci. Catal.* 130 (2000) 2669–2674.
- [12] C. Yang, Z. Meng, Bimolecular condensation of ethanol to 1-butanol catalyzed by alkali cation zeolites, *J. Catal.* 142 (1993) 37–44.
- [13] S. Ogo, A. Onda, K. Yanagisawa, Selective synthesis of 1-butanol from ethanol over strontium phosphate hydroxyapatite catalysts, *Appl. Catal. A Gen.* 402 (2011) 188–195, <http://dx.doi.org/10.1016/j.apcata.2011.06.006>.
- [14] K. Ramesh, E. Goh, Y. Ling, C.G. Gwie, T.J. White, A. Borgna, Structure and surface reactivity of WO₄²⁻, SO₄²⁻, PO₄³⁻ modified Ca-hydroxyapatite catalysts and their activity in ethanol conversion, *J. Phys. Chem. C* 116 (2012) 18736–18745.
- [15] J. Scalbert, F. Thibault-Starzyk, R. Jacquot, D. Morvan, F. Meunier, Ethanol condensation to butanol at high temperatures over a basic heterogeneous catalyst: how relevant is acetaldehyde self-aldolization?, *J. Catal.* 311 (2014) 28–32, <http://dx.doi.org/10.1016/j.jcat.2013.11.004>.
- [16] M.J.L. Gines, E. Iglesia, Bifunctional condensation reactions of alcohols on basic oxides modified by copper and potassium, *J. Catal.* 176 (1998) 155–172, <http://dx.doi.org/10.1006/jcat.1998.2009>.
- [17] T. Tsuchida, S. Sakuma, T. Takeguchi, W. Ueda, Direct synthesis of n-butanol from ethanol over nonstoichiometric hydroxyapatite, *Ind. Eng. Chem. Res.* 45 (2006) 8634–8642, <http://dx.doi.org/10.1021/ie0606082>.
- [18] T. Tsuchida, J. Kubo, T. Yoshioka, S. Sakuma, T. Takeguchi, W. Ueda, Reaction of ethanol over hydroxyapatite affected by Ca/P ratio of catalyst, *J. Catal.* 259 (2008) 183–189, <http://dx.doi.org/10.1016/j.jcat.2008.08.005>.
- [19] S. Hanspal, Z.D. Young, H. Shou, R.J. Davis, Multiproduct steady-state isotopic transient kinetic analysis of the ethanol coupling reaction over hydroxyapatite and magnesia, *ACS Catal.* 5 (2015) 1737–1746, <http://dx.doi.org/10.1021/cs502023g>.
- [20] I.M. Hill, S. Hanspal, Z.D. Young, R.J. Davis, DRIFTS of probe molecules adsorbed on magnesia, zirconia, and hydroxyapatite catalysts, *J. Phys. Chem. C* 119 (2015) 9186–9197, <http://dx.doi.org/10.1021/jp509889j>.
- [21] C.R. Ho, S. Shylesh, A.T. Bell, Mechanism and kinetics of ethanol coupling to butanol over hydroxyapatite, *ACS Catal.* 6 (2016) 939–948, <http://dx.doi.org/10.1021/acscatal.5b02672>.
- [22] S. Diallo-Garcia, M. Ben Osman, J. Kra, S. Casale, C. Thomas, J. Kubo, Identification of surface basic sites and acid–base pairs of hydroxyapatite, *J. Phys. Chem. C* 118 (2014) 12744–12757.
- [23] T. Tsuchida, J. Kubo, T. Yoshioka, S. Sakuma, T. Takeguchi, W. Ueda, Influence of preparation factors on Ca/P ratio and surface basicity of hydroxyapatite catalyst, *J. Japan. Pet. Inst.* 52 (2009) 51–59, <http://dx.doi.org/10.1627/jpi.52.51>.
- [24] I.R. Gibson, I. Rehman, S.M. Best, W. Bonfield, Characterization of the transformation from calcium-deficient apatite to β -tricalcium phosphate, *J. Mater. Sci. Mater. Med.* 11 (2000) 799–804, <http://dx.doi.org/10.1023/A:1008905613182>.
- [25] S. Lohrer, W.J. Stark, M. Maciejewski, A. Baiker, S.E. Pratsinis, D. Reichardt, F. Maspero, F. Krumeich, A. Biosciences, E.T.H. Hoenggerberg, C. Zurich, Fluoroapatite and calcium phosphate nanoparticles by flame synthesis, *Chem. Mater.* 17 (2005) 36–42.
- [26] R. Fabian, I. Kotsis, P. Zimany, P. Halmos, Preparation and chemical characterization of high purity fluorapatite, *Talanta* 46 (1998) 1273–1277.
- [27] J.C. Elliott, Structure and Chemistry of the Apatites and Other Calcium Orthophosphates, Elsevier Science & Technology, 1994.
- [28] S. Bordawekar, E. Dskocil, R. Davis, Microcalorimetric study of CO₂ and NH₃ adsorption on Rb- and Sr-modified catalyst supports, *Langmuir* 14 (1998) 1734–1738.
- [29] H. Tanaka, T. Watanabe, M. Chikazawa, FTIR and TPD studies on the adsorption of pyridine, n-butylamine and acetic acid on calcium hydroxyapatite, *J. Chem. Soc. Faraday Trans.* 93 (1997) 4377–4381.
- [30] D. Stošić, S. Bennici, S. Sirotni, C. Calais, J.-L. Couturier, J.-L. Dubois, A. Travert, A. Auroux, Glycerol dehydration over calcium phosphate catalysts: effect of acidic–basic features on catalytic performance, *Appl. Catal. A Gen.* 447 (2012) 124–134, <http://dx.doi.org/10.1016/j.apcata.2012.09.029>.
- [31] J.T. Kozlowski, R.J. Davis, Heterogeneous catalysts for the Guerbet coupling of alcohols, *ACS Catal.* 3 (2013) 1588–1600, <http://dx.doi.org/10.1021/cs400292f>.
- [32] R.L. Wingad, P.J. Gates, S.T.G. Street, D.F. Wass, Catalytic conversion of ethanol to n-butanol using ruthenium P–N ligand complexes, *ACS Catal.* 5 (2015) 5822–5826, <http://dx.doi.org/10.1021/acscatal.5b01327>.
- [33] T. Riittonen, E. Toukonitty, D.K. Madhani, A.-R. Leino, K. Kordas, M. Szabo, A. Sapi, K. Arve, J. Wärmä, J.-P. Mikkola, One-pot liquid-phase catalytic conversion of ethanol to 1-butanol over aluminium oxide—the effect of the active metal on the selectivity, *Catalysts* 2 (2012) 68–84, <http://dx.doi.org/10.3390/catal2010068>.
- [34] V. Bolis, C. Busco, M. Gianmario, L. Bertinetti, Y. Sakhno, P. Ugliengo, F. Chiatti, M. Corno, N. Roveri, Coordination chemistry of Ca sites at the surface of nanosized hydroxyapatite: interactions with H₂O and CO, *Philis. Trans. R. Soc. A* 370 (2012) 1313–1336.
- [35] B. Fubini, V. Bolis, M. Bailes, F.S. Stone, The reactivity of oxides with water vapor, *Solid State Ion.* 32–33 (1989) 258–272.
- [36] B.C. Gates, Catalytic Chemistry, John Wiley & Sons Inc., 1992.
- [37] M. Corno, C. Busco, V. Bolis, S. Tosoni, P. Ugliengo, Water adsorption on the stoichiometric (001) and (010) surfaces of hydroxyapatite: a periodic B3LYP study, *Langmuir* 25 (2009) 2188–2198, <http://dx.doi.org/10.1021/la803253k>.
- [38] Z.D. Young, S. Hanspal, R.J. Davis, Aldol condensation of acetaldehyde over titania, hydroxyapatite, and magnesia, *ACS Catal.* 6 (2016) 3193–3202, <http://dx.doi.org/10.1021/acscatal.6b00264>.
- [39] V.K. Díez, C.R. Apesteguía, J.I. Di Cosimo, Acid-base properties and active site requirements for elimination reactions on alkali-promoted MgO catalysts, *Catal. Today* 63 (2000) 53–62, [http://dx.doi.org/10.1016/S0920-5861\(00\)00445-4](http://dx.doi.org/10.1016/S0920-5861(00)00445-4).
- [40] M. León, E. Díaz, S. Ordóñez, Ethanol catalytic condensation over Mg–Al mixed oxides derived from hydrotalcites, *Catal. Today* 164 (2011) 436–442.
- [41] K. Tanabe, M. Misono, Y. Ono, H. Hattori, New Solid Acids and Bases, Elsevier Science BV, 1989.
- [42] E.P. Parry, An infrared study of pyridine adsorbed characterization of surface, *J. Catal.* 2 (1963) 371–379.